TUTORIAL 2 ISD

Matière: Information System Design **Level :** GLSI2	PRISTINI SCHOOL OF AI	**Academic Year:** 2026/2027 **A.Prof**: Ameni YOUSSFI

T n ° 2

Information System Design

Class Diagram

Exercise 1:

E-commerce Website: A website that allows people to buy and sell physical goods, services, and digital products over the internet from the safety of your home. Through an e-commerce website, a business can process orders, accept payments, manage shipping and logistics, and provide customer service.

Task:

Represent the use case diagram and the class diagram for this system.

Exercise 2:

The Minesweeper game is composed of a rectangular board, a timer, and a mine counter. The board is a grid of cells. At the beginning of the game, all cells are covered, and the mine counter shows the number of mines remaining to be located. The timer counts the seconds elapsed since the start of the game.

The game starts when the first cell is uncovered. When a cell is uncovered, its content is displayed. The content of a cell can be: nothing, a mine, or a number indicating the number of mines in neighboring cells.

The following scenarios can occur when a cell is uncovered, depending on its content:

1. A number – Nothing happens.
2. Blank – All neighboring cells are revealed, provided they are not flagged.

If one of these neighboring cells contains nothing, the uncovering process continues automatically from that cell.

3. A mine – The game ends and the player lose.

If a cell is still covered, it can be marked according to the following rules:

- Marking a cell that is neither uncovered nor already marked decrements the remaining mine counter, and a flag appears on the cell, indicating that the cell potentially contains a mine. A flagged cell cannot be uncovered.
- Marking a cell that is already flagged restores it to its initial state, i.e., covered and unmarked. The mine counter is then incremented by 1.

Task:

Represent the class diagram for this system.

Exercise3:

A computer team is composed of developers. The team works for a company and has an identifier and a logo. A developer is characterized by their ID number, last name, first name, and email. Each developer uses a personal computer.

A developer can be a programmer specialized in one or more programming languages, or a designer expert in a modeling method or language. Developers write code, while designers interact with users and create models.

Tasks:

1. Present the class diagram related to the description above.
2. Create an object diagram corresponding to the following text: Ali and Zied are programmers specialized in Ada and Java, respectively. Anis is a UML designer. They are part of the team “DEV”, which represents the company “TechSoft”. The logo of the DEV team is “DEV+”. All developers use the same computer.